MATH FOUNDATIONS

Probability Visual Guide

Bayes' theorem, distributions, and expectation — reasoning rigorously under uncertainty.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

The Three Axioms

Kolmogorov's rules — everything else is a consequence.

1. $0 \le P(A) \le 1$

probabilities live between zero and one

2. $P(\text{Sample space}) = 1$

something must happen

3. $P(A \text{ or } B) = P(A) + P(B)$

for disjoint A, B — probabilities add

These three lines imply every formula you will meet: complements, unions, conditionals.

STEP 2/6

Conditional Probability & Independence

New evidence shrinks the sample space to what you observed.

P(A | B) = P(A and B) / P(B)

'probability of A GIVEN that B happened'

INDEPENDENT

P(A|B) = P(A) — B tells you nothing

DEPENDENT

P(A|B) != P(A) — B changes the odds

Assuming independence where none exists is the most common modeling sin — and the most expensive.

STEP 3/6

Bayes' Theorem: Flip the Conditional

From P(evidence | hypothesis) to P(hypothesis | evidence).

P(H | E) = P(E | H) * P(H) / P(E)

posterior = likelihood * prior / evidence

Rare diseases, spam filters, sensor fusion: Bayes is how evidence updates belief, correctly.

STEP 4/6

Three Distributions to Know Cold

Each answers a different 'how many / how often' question.

BINOMIAL

k successes in n trials: coin flips, A/B conversions

POISSON

events per interval: arrivals/sec, errors per deploy

NORMAL (Gaussian)

bell curve: measurement noise, sums of many small effects

The normal distribution earns its fame from the Central Limit Theorem — averages become normal as n grows.

STEP 5/6

Expectation & Variance

The center of mass of a distribution, and how spread out it is.

E[X] = sum x*P(x)

long-run average value

Var(X) = E[(X-mu)^2]

average squared deviation

Linearity: E[aX + bY] = aE[X] + bE[Y] — works even when X, Y are dependent.

STD DEV = sqrt(Var)

same units as X — the honest error bar on any estimate

Means deceive; variance tells you whether the mean is a promise or a rumor.

STEP 6/6

LLN and CLT: Why Statistics Works

Two theorems that make measurement meaningful.

LAW OF LARGE NUMBERS

sample mean -> true mean as n grows. More data = closer truth.

CENTRAL LIMIT THEOREM

means of samples become normal — whatever the original shape.

CLT is why confidence intervals and A/B test math work even when your data is wildly non-normal.

STEP ✓I✓

Key Takeaways

Probability — Remember

- Three axioms (bounded, total = 1, additive) imply the entire theory.
- Conditioning shrinks the world to what you observed; check independence assumptions ruthlessly.
- Bayes flips the conditional: posterior = likelihood x prior / evidence.
- Binomial counts successes in trials; Poisson counts events per interval; normal models noise.
- Expectation is the long-run average; variance is the honest error bar.
- LLN says more data approaches truth; CLT says sample means become normal.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev